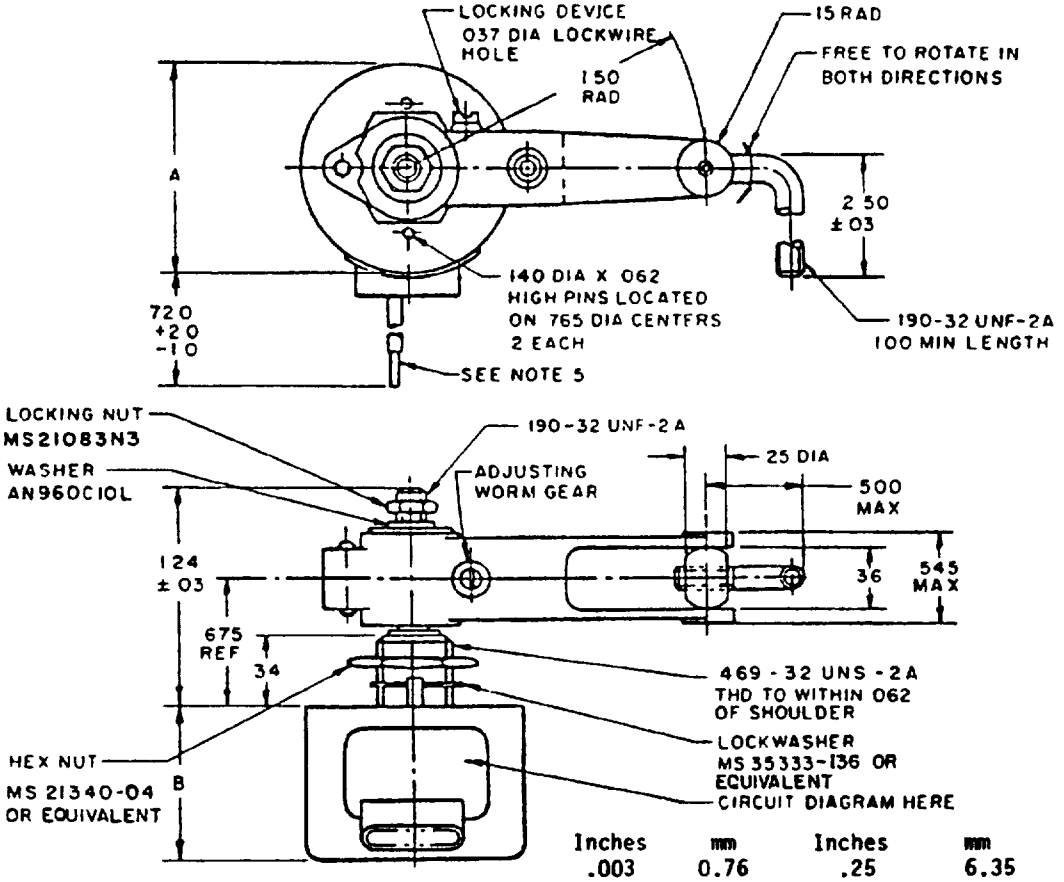
MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, LIMIT, ROTARY LINKAGE LEVER (NONSELF-RETURN), RESILIENT SEAL, FLUID RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

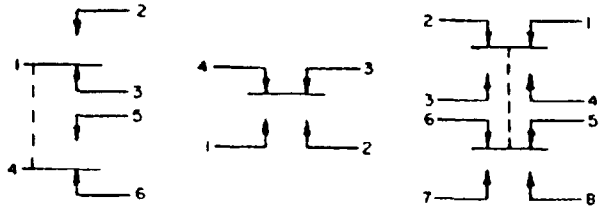


Inches	mm	Inches	mm	Inches	mm
.003	0.76	.25	6.35	.765	19.43
.037	0.94	.34	8.64	1.0	25.4
.062	1.57	.36	9.14	1.24	31.50
.140	3.56	.469	11.91	1.50	38.10
.15	3.8	.500	12.70	2.0	50.
.190	4.83	.545	13.84	2.50	63.5
		.675	17.15	72.0	1828.

FIGURE 1. Configuration and dimensions.

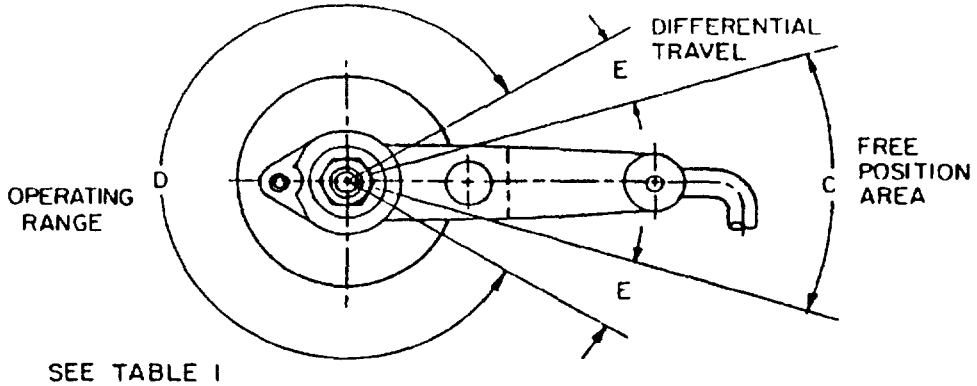
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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DPDT 2 CIRCUIT 4 CIRCUIT

CIRCUIT SCHEMATICS



SEE TABLE I

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (0.51 mm) for two place decimals and ±.005 (0.13 mm) for three place decimals.
4 Lever shall be adjustable for an infinite number of positions through 360° without removing lever.
5. Leads shall be marked in accordance with MIL-W-5088 with the switch circuit identification number followed by wire gauge number (1-20, 2-20, etc). Lead wire shall be MIL-W-22759/7
6 Envelope optional provided dimensions specified are not exceeded.

FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS

Dimensions and configuration See figure 1 and table I

TABLE I Dimensions and operating characteristics

MS24420-	Maximum dimensions		Wire leads		Operating characteristics 1/					
					C		D		E	
	A	B	Wire size	Number required	Free position area	Circuits made	D Operating range	Circuits made	Maximum differential travel	Maximum operating torque inch-pounds
1	1.030 (26.16)	1.050 (26.67)	20	6	35° ±15°	1-3 and 4-6	301° REF	1-2 and 4-5	12°	3
2		1.680 (42.67)								5
3	1.530 (38.86)	1.430 (36.32)	18	4		3 and 4	285° REF	1 and 2	20°	10
4		1.820 (46.23)		8		1-2 and 5-6		3-4 and 7-8		
5			20							

1/ ±20 percent variation from specified values acceptable after tests.

Enclosure design: Four (resilient). All entrances to the switch cavity except through the actuator bushing shall be sealed by fusion of glass-to-metal, metal-to-metal, or ceramic-to-metal and the lead wires shall be potted to provide stress relief.

Temperature characteristic: One (-55°C to +85°C).

Shock type: M (100 G).

Sinusoidal vibration grade: One (10 to 500 Hz).

Material.

Plastic:

MS24420-1: Type CFG in accordance with MIL-M-14 may be used internally.

MS24420-2: General purpose mineral and flock-filled phenolics may be used internally.

Finish: Switch housing shall be processed to resist corrosion.

Maximum weight with leads:

MS24420-1: .52 pound.

MS24420-2: .75 pound.

MS24420-3. .61 pound.

MS24420-4, -5: 1.10 pounds.

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Operating characteristics See table I

Coincidence of operating and releasing point Not applicable

Contact resistance. Not applicable

Insulation resistance. 100 meghoms minimum

Dielectric withstanding voltage

At atmospheric pressure 1,000 V rms

At reduced barometric pressure 50,000 feet, 400 V rms

Mechanical endurance 25,000 cycles.

Electrical endurance. 25,000 cycles.

Electrical ratings: See table II.

TABLE II. Electrical ratings.

Rating code	Load							
	Sea level - 28 V dc				50,000 feet - 28 V dc			
	Resistive	Inductive	Lamp	Motor	Resistive	Inductive	Lamp	Motor 1/
	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)
A	4	3	2.4	4	4	1	2.4	4
B	10	6	3.6	6	10	3	3.6	6
C	15	10	3.0	5	15	10	3.0	5

1/ Application information only.

Fluid resistance: Except for the cut end of the lead wire, switches shall be submerged in each of the following fluids for 2 minutes to 2 minutes 30 seconds, which shall consist of one cycle (one cycle is 10 minutes to 12 minutes 30 seconds total). Each switch shall be subjected to three cycles.

a. Turbine fuel (MIL-T-5624).

b Hydraulic fluid (SAE AS1241A).

c. Coolanol 1/ (MIL-C-47220).

d Ethylene glycol (MIL-E-9500).

e. Lubricating oil (MIL-L-7808).

After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for dielectric withstanding voltage, insulation resistance, operating characteristics, seal tests, and marking visibility.

Marking. The circuit schematic shall be marked on the switch case.

1/ Monsanto Company registered trademark.

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Part or Identifying Number (PIN) See table III

TABLE III. PIN and characteristics.

PIN	Electrical rating code (table II)	Circuit
MS24420-1	A	DPDT
MS24420-2	B	DPDT
MS24420-3	C	2 circuit
MS24420-4	C	4 circuit
MS24420-5	B	4 circuit

Qualification inspection

Group submission. See table IV.

Group A inspection

Seal test. Only watertight test shall be performed.

TABLE IV. Qualification inspection (group submission).

Examination or test	Samples	Extent of approval
Qualification inspection table of MIL-S-8805	MS24420-4 (28 units)	All
Visual and mechanical examination, group II	MS24420-1, -3 (2 units each)	
Group VII	MS24420-1, -2 (14 units each)	

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS
Air Force - 99
DLA - ES

User activities.
Army - AV, MI
Navy - MC
Air Force - 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5930-1413-04)

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